

Questions with Answer Keys

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Q1 - Single Correct

If $\vec{a}, \vec{b}$ are unit vectors such that $(\vec{a} + \vec{b}) \cdot \{(2\vec{a} + 3\vec{b}) \times (3\vec{a} - 2\vec{b})\} = 0$, then angle between $\vec{a}$ and $\vec{b}$ is -

- (1) 0
- (2) $\pi/2$
- (3) π
- (4) indeterminate

Q2 - Single Correct

If $\vec{a}' = \hat{i} + \hat{j}$, $\vec{b}' = \hat{i} - \hat{j} + 2\hat{k}$ & $\vec{c}' = 2\hat{i} + \hat{j} - \hat{k}$. Then altitude of the parallelopiped formed by the vectors $\vec{a}, \vec{b}, \vec{c}$ having base formed by $\vec{b}$ & $\vec{c}$ is ($\vec{a}, \vec{b}, \vec{c}$ and $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal systems of vectors)

- (1) 1
- (2) $\frac{3\sqrt{2}}{2}$
- (3) $\frac{1}{\sqrt{6}}$
- (4) $\frac{1}{\sqrt{2}}$

Q3 - Single Correct

A unit vector $\vec{a}$ in the plane of $\vec{b} = 2\hat{i} + \hat{j}$ & $\vec{c} = \hat{i} - \hat{j} + \hat{k}$ is such that $\vec{a} \wedge \vec{b} = \vec{a} \wedge \vec{c}$ where $\vec{d} = \hat{j} + 2\hat{k}$ is

- (1) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
- (2) $\frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$
- (3) $\frac{2\hat{i} + \hat{j}}{\sqrt{5}}$
- (4) $\frac{2\hat{i} + \hat{j}}{\sqrt{5}}$

Q4 - Single Correct

If $\vec{a}_1, \vec{a}_2$ & $\vec{a}_3$ are non-coplanar vectors and $(x + y - 3)\vec{a}_1 + (2x - y + 2)\vec{a}_2 + (2x + y + \lambda)\vec{a}_3 = \vec{0}$ holds for some 'x' and 'y' then ' λ ' is

- (1) $\frac{7}{3}$

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(2) 2

(3) $-\frac{10}{3}$

(4) $\frac{5}{3}$

Q5 - Single Correct

If $\vec{a}, \vec{b}, \vec{c}$ are such that $[\vec{a}, \vec{b}, \vec{c}] = 1, \vec{c} = \lambda \vec{a} \times \vec{b}, \vec{a} \wedge \vec{b} < \frac{2\pi}{3}$ and $|\vec{a}| = \sqrt{2}, |\vec{b}| = \sqrt{3}, |\vec{c}| = \frac{1}{\sqrt{3}}$, then the angle between $\vec{a}$ & $\vec{b}$ is

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{3}$

(4) $\frac{\pi}{2}$

Q6 - Single Correct

If the plane $\vec{r} \cdot (4\hat{i} + 3\hat{j}) + 2 = 0$ is rotated about its line of intersection with the plane $\vec{r} \cdot (\hat{k}) = 0$ through an angle 45° , then the length of perpendicular from the origin on the new position of the plane is -

(1) $\frac{1}{5}$

(2) $\frac{2}{5}$

(3) $\frac{\sqrt{2}}{5}$

(4) $\frac{\sqrt{3}}{5}$

Q7 - Single Correct

In a right angled $\triangle ABC$, the hypotenuse $AB = p$, then $\vec{AB} \cdot \vec{AC} + \vec{BC} \cdot \vec{BA} + \vec{CA} \cdot \vec{CB}$ equals

(1) $2p^2$

(2) p^2

(3) $p^2/2$

(4) none of these

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Q8 - Single Correct

Equation of the straight line in the plane $\vec{r} \cdot \vec{n} = d$ which is parallel to $\vec{r} = \vec{a} + \lambda \vec{b}$ and passes through the foot of perpendicular drawn from the point $P(\vec{a})$ to the plane $\vec{r} \cdot \vec{n} = d$ is (where $\vec{n} \cdot \vec{b} = 0$)

(1) $\vec{r} = \vec{a} + \left(\frac{d - \vec{a} \cdot \vec{n}}{\vec{n}^2} \right) \vec{n} + \lambda \vec{b}$

(2) $\vec{r} = \vec{a} + \left(\frac{d - \vec{a} \cdot \vec{n}}{\vec{n}} \right) \vec{n} + \lambda \vec{b}$

(3) $\vec{r} = \vec{a} + \left(\frac{\vec{a} \cdot \vec{n} - d}{\vec{n}^2} \right) \vec{n} + \lambda \vec{b}$

(4) $\vec{r} = \vec{a} + \left(\frac{\vec{a} \cdot \vec{n} - d}{\vec{n}} \right) \vec{n} + \lambda \vec{b}$

Q9 - Single Correct

Equation of line in the plane $\pi \equiv 2x - y + z - 4 = 0$ which is perpendicular to the line ℓ whose equation is $\frac{x-2}{1} = \frac{y-2}{-1} = \frac{z-3}{-2}$ and which passes through the point of intersection of ℓ and π is

(1) $\frac{x-2}{3} = \frac{y-1}{5} = \frac{z-1}{-1}$

(2) $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-5}{-1}$

(3) $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z+1}{1}$

(4) $\frac{x-2}{2} = \frac{y-1}{-1} = \frac{z-1}{1}$

Q10 - Single Correct

Equation of plane which passes through the point of intersection of lines $\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and

$\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and at greatest distance from the point $(0, 0, 0)$ is:

(1) $4x + 3y + 5z = 25$

(2) $4x + 3y + 5z = 50$

(3) $3x + 4y + 5z = 49$

(4) $x + 7y - 5z = 2$

Q11 - Single Correct

Let $A(\vec{a})$ and $B(\vec{b})$ be points on two skew lines $\vec{r} = \vec{a} + \lambda \vec{p}$ and $\vec{r} = \vec{b} + u \vec{q}$ and the shortest distance between the skew lines is 1, where $\vec{p}$ and $\vec{q}$ are unit vectors forming adjacent sides of a parallelogram

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enclosing an area of $\frac{1}{2}$ units. If an angle between AB and the line of shortest distance is 60° , then AB =

- (1) $\frac{1}{2}$
- (2) 2
- (3) 1
- (4) 5

Q12 - Single Correct

If $P_1 : \vec{r} \cdot \vec{n}_1 - d_1 = 0$, $P_2 : \vec{r} \cdot \vec{n}_2 - d_2 = 0$ and $P_3 : \vec{r} \cdot \vec{n}_3 - d_3 = 0$ are three planes and $\vec{n}_1, \vec{n}_2$ and $\vec{n}_3$ are three non-coplanar vectors then, the three lines $P_1 = 0, P_2 = 0$; $P_2 = 0, P_3 = 0$ and $P_3 = 0, P_1 = 0$ are

- (1) parallel lines
- (2) coplanar lines
- (3) coincident lines
- (4) concurrent lines

Q13 - Single Correct

If $\vec{a}, \vec{b}$ and $\vec{c}$ are three unit vectors equally inclined to each other at an angle α . Then the angle between $\vec{a}$ and plane of $\vec{b}$ and $\vec{c}$ is

- (1) $\theta = \cos^{-1} \left(\frac{\cos \alpha}{\cos \frac{\alpha}{2}} \right)$
- (2) $\theta = \sin^{-1} \left(\frac{\cos \alpha}{\cos \frac{\alpha}{2}} \right)$
- (3) $\theta = \cos^{-1} \left(\frac{\sin \frac{\alpha}{2}}{\sin \alpha} \right)$
- (4) $\theta = \sin^{-1} \left(\frac{\sin \frac{\alpha}{2}}{\sin \alpha} \right)$

Q14 - Single Correct

Let A(1, 1, 1), B(2, 3, 5), C(-1, 0, 2) be three points, then equation of a plane parallel to the plane ABC which is at a distance 2 from origin, is

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$$(1) 2x - 3y + z + 2\sqrt{14} = 0$$

$$(2) 2x - 3y + z - \sqrt{14} = 0$$

$$(3) 2x - 3y + z + 2 = 0$$

$$(4) 2x - 3y + z - 2 = 0$$

Q15 - Multiple Correct

If $\vec{a}$ and $\vec{b}$ unequal unit vectors such that $(\vec{a} - \vec{b}) \times [(\vec{b} + \vec{a}) \times (2\vec{a} + \vec{b})] = \vec{a} + \vec{b}$, then smaller angle θ between $\vec{a}$ and $\vec{b}$ is

$$(1) \frac{\pi}{2}$$

$$(2) 0$$

$$(3) \pi$$

$$(4) \frac{\pi}{4}$$

Q16 - Multiple Correct

In a four - dimensional space where unit vectors along axes are $\hat{i}, \hat{j}, \hat{k}$ and $\hat{\ell}$ and $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ are four non-zero vectors such that no vector can be expressed as linear combination of others and

$$(\lambda - 1)(\vec{a}_1 - \vec{a}_2) + \mu(\vec{a}_2 + \vec{a}_3) + \gamma(\vec{a}_3 + \vec{a}_4 - 2\vec{a}_2) + \vec{a}_3 + \delta\vec{a}_4 = \vec{0} \text{ then}$$

$$(1) \lambda = 1,$$

$$(2) \mu = -\frac{2}{3}$$

$$(3) \gamma = \frac{2}{3}$$

$$(4) \delta = \frac{1}{3}$$

Q17 - Multiple Correct

A vector ($\vec{d}$) is equally inclined to three vectors $\vec{a} = \hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{j}$ and $\vec{c} = 3\hat{j} - 2\hat{k}$. Let

$\vec{x}, \vec{y}, \vec{z}$ be three vector in the plane of $\vec{a}, \vec{b}; \vec{b}, \vec{c}; \vec{c}, \vec{a}$ respectively, then

$$(1) \vec{x} \cdot \vec{d} = 14$$

$$(2) \vec{y} \cdot \vec{d} = 3$$

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(3) $\vec{z} \cdot \vec{d} = 0$

(4) $\vec{r} \cdot \vec{d} = 0$ where $\vec{r} = \lambda\vec{x} + \mu\vec{y} + \delta\vec{z}$

Q18 - Multiple Correct

Identify the statement(s) which is/are INCORRECT ?

(1) $\vec{a} \times [\vec{a} \times (\vec{a} \times \vec{b})] = (\vec{a} \times \vec{b}) (\vec{a}^2)$

(2) If $\vec{a}, \vec{b}, \vec{c}$ are non-zero, non coplanar vectors, and $\vec{v} \cdot \vec{a} = \vec{v} \cdot \vec{b} = \vec{v} \cdot \vec{c} = 0$ then $\vec{v}$ must be a null vector

(3) If $\vec{a}$ and $\vec{b}$ lie in a plane normal to the plane containing the vectors $\vec{a} \times \vec{b}, \vec{c} \times \vec{d}$; $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are non-zero vector, then $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$

(4) If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{a}', \vec{b}', \vec{c}'$ are reciprocal system of vectors then $\vec{a} \cdot \vec{b}' + \vec{b} \cdot \vec{c}' + \vec{c} \cdot \vec{a}' = 3$

Q19 - Multiple Correct

$\vec{a}, \vec{b}, \vec{c}$ are three co-planar unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. If three vectors $\vec{p}, \vec{q}, \vec{r}$ parallel to $\vec{a}, \vec{b}, \vec{c}$

respectively and having integral but different magnitudes, then among the following options $|\vec{p} + \vec{q} + \vec{r}|$

can take a value equal to

(1) 1

(2) 0

(3) $\sqrt{3}$

(4) 2

Q20 - Multiple Correct

The equation of the line $x + y + z - 1 = 0, 4x + y - 2z + 2 = 0$ written in the symmetrical form is

(1) $\frac{x+1}{1} = \frac{y-2}{-2} = \frac{z-0}{1}$

(2) $\frac{x}{1} = \frac{y}{-2} = \frac{z-1}{1}$

(3) $\frac{x+1/2}{1} = \frac{y-1}{-2} = \frac{z-1/2}{1}$

(4) $\frac{x-1}{2} = \frac{y+2}{-1} = \frac{z-2}{2}$

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Q21 - Multiple Correct

Consider the planes $3x - 6y + 2z + 5 = 0$ and $4x - 12y + 3z = 3$. The plane $67x - 162y + 47z + 44 = 0$ bisects that angle between the given planes which

- (1) contains origin
- (2) is acute
- (3) is obtuse
- (4) none of these

Q22 - Multiple Correct

The plane $\ell x + my = 0$ is rotated about its line of intersection with the plane $z = 0$, through an angle α , then equation of plane in its new position may be

- (1) $\ell x + my + z\sqrt{\ell^2 + m^2} \tan \alpha = 0$
- (2) $\ell x + my - z\sqrt{\ell^2 + m^2} \tan \alpha = 0$
- (3) data is not sufficient
- (4) None of these

Q23 - Paragraph 1

Passage I (For Question 23, 24)

Consider two lines $L_1 : \frac{x-2}{1} = \frac{y-1}{0} = \frac{z+1}{2}$ and $L_2 : \frac{x-3}{1} = \frac{y-1}{1} = \frac{z-0}{-1}$. Let P be the plane which contains the line L_1 and is parallel to L_2 and intersecting coordinates axes at A, B, C respectively -

The shortest distance between the lines L_1 and L_2 is -

- (1) $\frac{1}{\sqrt{5}}$
- (2) $\frac{1}{\sqrt{6}}$
- (3) $\frac{1}{\sqrt{8}}$
- (4) $\frac{1}{\sqrt{14}}$

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Q24 - Paragraph 1

Image of origin in the plane P , is -

- (1) $\left(\frac{2}{7}, \frac{-3}{7}, \frac{-1}{7}\right)$
- (2) $\left(\frac{4}{7}, \frac{-6}{7}, \frac{-2}{7}\right)$
- (3) $\left(\frac{-2}{7}, \frac{3}{7}, \frac{1}{7}\right)$
- (4) $\left(\frac{-4}{7}, \frac{6}{7}, \frac{2}{7}\right)$

Q25 - Paragraph 2

Passage II (For Question 26, 27)

Suppose direction cosines of two lines are given by $u\ell + vm + wn = 0$ and $a\ell^2 + bm^2 + cn^2 = 0$, where u, v, w, a, b, c are arbitrary constants and ℓ, m, n are direction cosines of the lines.

For $u = v = w = 1$, both lines satisfies the relation -

- (1) $(b + c)\left(\frac{n}{\ell}\right)^2 + 2b\frac{n}{\ell} + (a + b) = 0$
- (2) $(c + a)\left(\frac{\ell}{m}\right)^2 + 2c\frac{\ell}{m} + (b + c) = 0$
- (3) $(a + b)\left(\frac{m}{n}\right)^2 + 2a\frac{m}{n} + (c + a) = 0$
- (4) All of the above

Q26 - Paragraph 2

For $u = v = w = 1$, If $\frac{n_1 n_2}{\ell_1 \ell_2} = \frac{a+b}{b+c}$, then

- (1) $\frac{m_1 m_2}{\ell_1 \ell_2} = \frac{b+c}{c+a}$
- (2) $\frac{m_1 m_2}{\ell_1 \ell_2} = \frac{c+a}{b+c}$
- (3) $\frac{m_1 m_2}{\ell_1 \ell_2} = \frac{a+b}{c+a}$
- (4) $\frac{m_1 m_2}{\ell_1 \ell_2} = \frac{c+a}{a+b}$

Q27 - Paragraph 3

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Passage III (For Question 27, 28)

Let $\vec{a} = 2\hat{i} + 3\hat{j} - 6\hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ and $\vec{c} = -2\hat{i} + 3\hat{j} + 6\hat{k}$. Let $\vec{a}_1$ be projection of $\vec{a}$ on $\vec{b}$ and $\vec{a}_2$ be the projection of $\vec{a}_1$ on $\vec{c}$, then

$$\vec{a}_2 =$$

(1) $\frac{943}{49}(2\hat{i} - 3\hat{j} - 6\hat{k})$

(2) $\frac{943}{49^2}(2\hat{i} - 3\hat{j} - 6\hat{k})$

(3) $\frac{943}{49}(-2\hat{i} + 3\hat{j} + 6\hat{k})$

(4) $\frac{943}{49^2}(-2\hat{i} + 3\hat{j} + 6\hat{k})$

Q28 - Paragraph 3

$$\vec{a}_1 \cdot \vec{b} =$$

(1) -41

(2) $-\frac{41}{7}$

(3) 41

(4) 287

Q29 - Paragraph 4**Passage IV (For Question 29, 30)**

ABCD is a parallelogram L is a point on BC which divides BC in the ratio 1 : 2. AL intersects BD at P. M is a point on DC which divides DC in the ratio 1 : 2 and AM intersects BD in Q.

Point P divides AL in the ratio

(1) 1 : 2

(2) 1 : 3

(3) 3 : 1

(4) 2 : 1

Questions with Answer Keys

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Q30 - Paragraph 4

Point Q divides DB in the ratio

- (1) $1 : 2$
- (2) $1 : 3$
- (3) $3 : 1$
- (4) $2 : 1$

Q31 - Paragraph 5

Passage V (For Question 31, 32)

Three vector $\hat{a}$, $\hat{b}$ and $\hat{c}$ are forming a right handed system, if $\hat{a} \times \hat{b} = \hat{c}$, $\hat{b} \times \hat{c} = \hat{a}$, $\hat{c} \times \hat{a} = \hat{b}$, then answer the following question

If vector $3\hat{a} - 2\hat{b} + 2\hat{c}$ and $-\hat{a} - 2\hat{c}$ are adjacent sides of a parallelogram, then an angle between the diagonals is

- (1) $\frac{\pi}{4}$
- (2) $\frac{\pi}{3}$
- (3) $\frac{\pi}{2}$
- (4) $\frac{2\pi}{3}$

Q32 - Paragraph 5

If $\vec{x} = \hat{a} + \hat{b} - \hat{c}$, $\vec{y} = -\hat{a} + \hat{b} - 2\hat{c}$, $\vec{z} = -\hat{a} + 2\hat{b} - \hat{c}$, then a unit vector normal to the vectors $\vec{x} + \vec{y}$ and $\vec{y} + \vec{z}$ is

- (1) $\vec{a}$
- (2) $\vec{b}$
- (3) $\vec{c}$
- (4) none of these

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Q33 - Integer Type

If equation of the plane through the straight line $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$ and perpendicular to the plane $x - y + z + 2 = 0$ is $ax - by + cz + 4 = 0$, then find the value of $a + b + c$

Q34 - Integer Type

The shortest distance between the lines given by $\vec{r} = 3\vec{i} + 8\vec{j} + 3\vec{k} + \lambda(3\vec{i} - \vec{j} + \vec{k})$ and $\vec{r} = -3\vec{i} - 7\vec{j} + 6\vec{k} + \mu(-3\vec{i} + 2\vec{j} + 4\vec{k})$ is $\lambda\sqrt{30}$, then find the value of λ

Q35 - Integer Type

If V_1, V_2, V_3 are volumes of parallelopiped, triangular prism and tetrahedron respectively. The three coterminal edges of all three figures are the vectors $\hat{i} - \hat{j} - 6\hat{k}, \hat{i} - \hat{j} + 4\hat{k}$ and $2\hat{i} - 5\hat{j} + 3\hat{k}$, then find sum of the volumes V_1, V_2 and V_3

Q36 - Integer Type

If $\vec{p}, \vec{q}, \vec{r}$ are mutually perpendicular vectors, where $|\vec{p}| = |\vec{q}| = |\vec{r}|$ and $\vec{p} \times \{(\vec{x} - \vec{q}) \times \vec{p}\} + \vec{q} \times \{(\vec{x} - \vec{r}) \times \vec{q}\} + \vec{r} \times \{(\vec{x} - \vec{p}) \times \vec{r}\} = \vec{0}$ and $\vec{x} = \frac{1}{\lambda}(\vec{p} + \vec{q} + \vec{r})$, then find the value of λ .

Q37 - Integer Type

In a regular tetrahedron let θ be the angle between any edge and a face not containing the edge. If $\cos^2 \theta = \frac{a}{b}$ where $a, b \in I^+$ also a and b are coprime, then find the value of $10a + b$

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